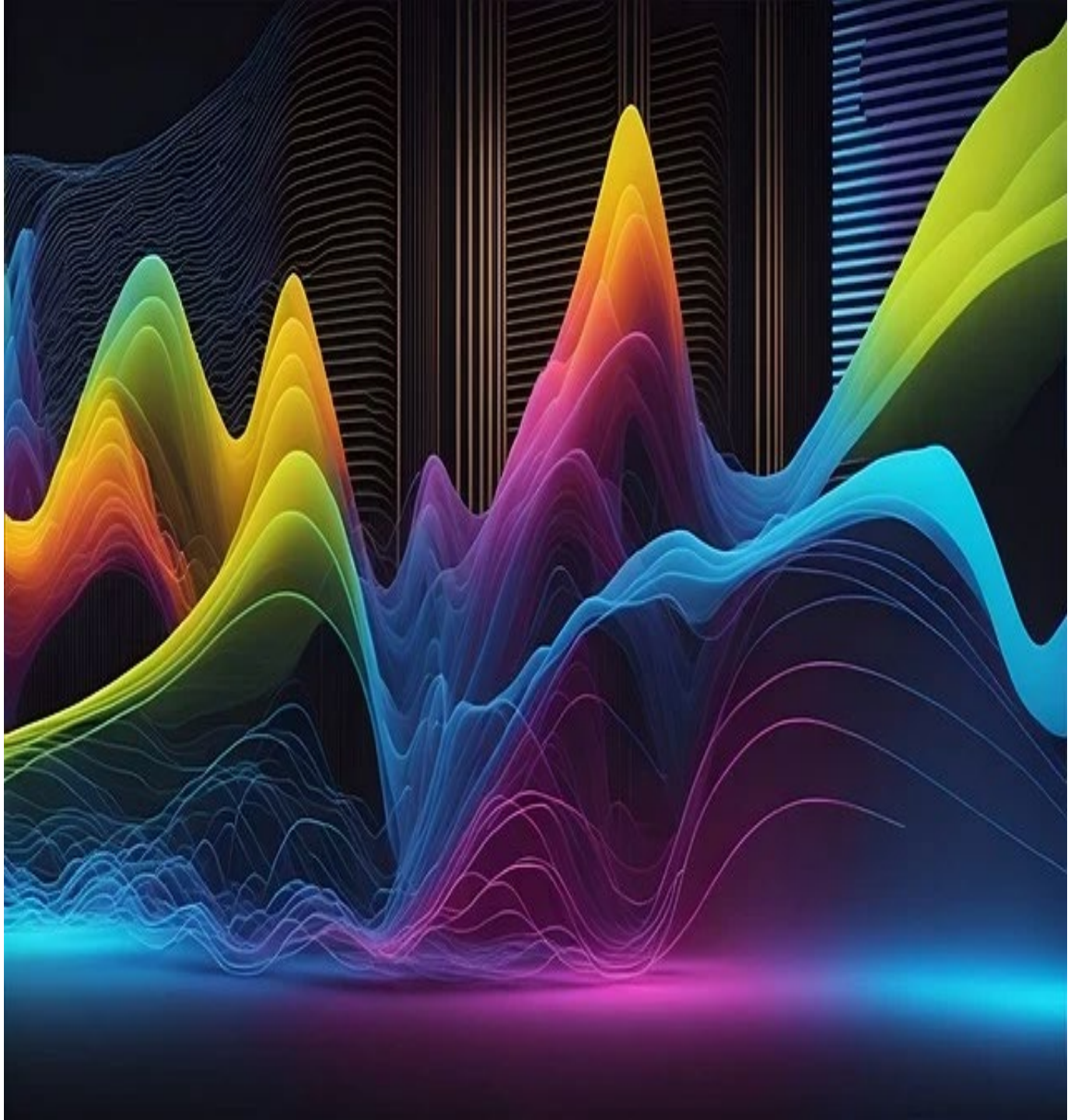
ME551 Vibrations in Linear Systems

Lecture 10

Multiple Degree of Freedom Systems 2 (Stability)



Chap 4. Multi-Degree-of-Freedom System

4.1 Equations of Motion
4.2 Geometric Representation of Motion. Equilibrium Points
4.3 Stability of Equilibrium Points. Liapunov Direct Method
4.4 Linearization about Equilibrium Points
4.5 Stability of Linearized Systems
4.6 Linear Conservative Natural Systems. Symmetric Eigenvalue Problem
4.7 Gyroscopic Conservative Systems
4.8 Nonconservative Systems. Nonsymmetric Eigenvalue Problem
4.9 Response of Discrete Systems to Harmonic Excitations
4.10 Response of Multi-DOF Systems to Arbitrary Excitations
i. Undamped natural systems
ii. Systems with proportional viscous damping
iii. Systems with small damping
iv. Systems with structural damping
v. Undamped gyroscopic systems
vi. General nonconservative systems
4.11 Discrete-Time Systems

 © Prof. Yong-Hwa Park, KAIST

4.3 Stability of Equilibrium Points. Liapunov Direct Method

- Remarks on **Stability**
  - ✓ Stability properties characterize the system behavior if its initial state is close to, but not at, an **equilibrium point** ($x_0 = [q_0^T, 0^T]^T$)of interest.
  - ✓ When an initial state is close to the equilibrium point, the state may remain close to (stable) or move away (unstable) from the equilibrium point.
  - ✓ An **equilibrium state** is **stable** if, whenever the initial state is near that point, the state <u>remains near</u> or <u>tends toward the equilibrium point as time increases</u>.

© Prof. Yong-Hwa Park, KAIST

Definitions of Stability by *Liapunov*

- An equilibrium state x_0 is **stable** if there exist a positive scalar $\delta = \delta(t_0, \varepsilon)$ such that if $\|x(t_0) - x_0\| < \delta$ then $\|x(t) - x_0\| < \varepsilon$ for $t \geq t_0$ for any given t_0 and positive scalar ε.

- The equilibrium state x_0 is **asymptotically stable** if it is stable and $\lim_{t \to \infty} \|x(t) - x_0\| = 0$.

- The equilibrium state x_0 is **unstable** if $\|x(t_1) - x_0\| = \varepsilon$ for a $t_1 > t_0$.

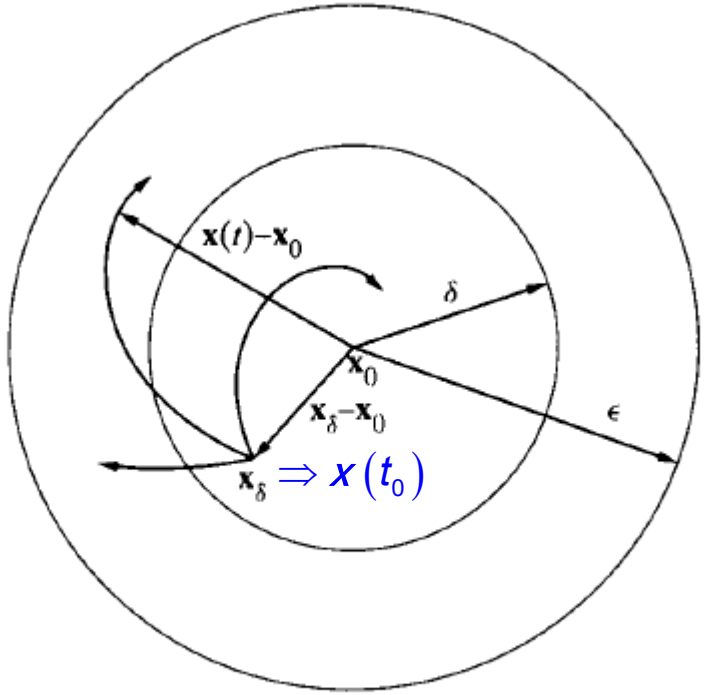


Figure 4.5 Stable, asymptotically stable and unstable trajectories in the state space

© Prof. Yong-Hwa Park, KAIST

The Positive (Negative) Definite of a Scalar Function

- The scalar function $\mathcal{V}(\boldsymbol{x})$ of state vector $\boldsymbol{x}$ is said to be **positive (negative) definite** if $\mathcal{V}(\boldsymbol{x}) > 0$ (< 0) for all $\boldsymbol{x} \neq 0$.

- The scalar function $\mathcal{V}(\boldsymbol{x})$ of state vector $\boldsymbol{x}$ is said to be **positive (negative) semidefinite** if $\mathcal{V}(\boldsymbol{x}) \geq 0$ (≤ 0) for all $\boldsymbol{x} \neq 0$.

- The scalar function $\mathcal{V}(\boldsymbol{x})$ of state vector $\boldsymbol{x}$ is said to be **sign-variable** if it can be both positive and negative values for $\boldsymbol{x} \neq 0$.

Ex) $\mathcal{V}(x_1, x_2) = \begin{cases} a_1 x_1^2 + a_2 x_2^2 \\ a_1 (x_1 - x_2)^2 \end{cases}$, where a_1 & $a_2 > 0$

© Prof. Yong-Hwa Park, KAIST

The *Liapunov* Direct Method: *Liapunov function*

- Procedure (when solutions are not available)

- Generation of a suitable <u>scalar testing function</u>, so called **Liapunov function** $\mathcal{V}(x)$, based on a rational, which may <u>not be unique</u>.
- Application of the **theorems** to the Liapunov function
- Even if a $\mathcal{V}(x)$ cannot be found, it doesn't mean that none of the three motion characteristics exist.

Ex) $\mathcal{V}(x) = \frac{1}{2} x^T A x, A$ is a positive definite matrix

== Eigenvalues of A are all positive. (EVP: $AX = \lambda X$)
== all the principal minor determinants of A is positive.

$$\det[A_{rs}] > 0, r, s = 1, 2, \ldots k; k = 1, 2, \ldots, 2n$$

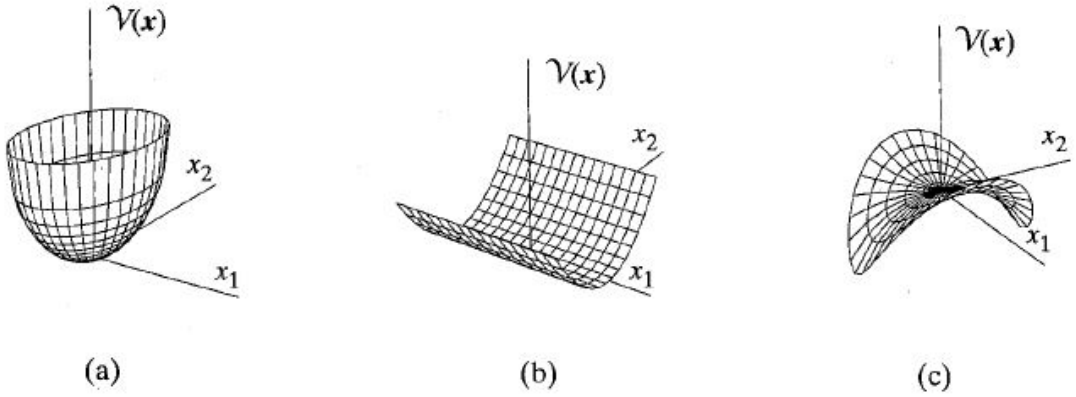


Figure 4.6 **(a)** Positive definite function **(b)** Positive semidefinite function **(c)** Sign-variable function

© Prof. Yong-Hwa Park, KAIST

Liapunov Stability Theorems for a System described by $\dot{x} = f(x)$

- **Theorem 1.** <u>If there exists</u> a **positive definite** function $\mathcal{V}(x)$ whose total time derivative $\dot{\mathcal{V}} = \sum_{i=1}^{2n} \frac{\partial \mathcal{V}}{\partial x_i} \dot{x}_i = \dot{x}^T \nabla \mathcal{V} = f^T \nabla \mathcal{V}$ is **negative semi-definite**, then an equilibrium state x_o which makes $\left(\mathcal{V} \cdot \dot{\mathcal{V}} \le 0\right)$ is **stable.**

- **Theorem 2.** If there exists a **positive definite** function $\mathcal{V}(x)$ whose total time derivative $\dot{\mathcal{V}}(x)$ is **negative definite**, then the equilibrium state x_o is **asymptotically stable.** $\left(\mathcal{V} \cdot \dot{\mathcal{V}} < 0\right)$

- **Theorem 3.** If there exists a function $\mathcal{V}(x)$ whose total time derivative $\dot{\mathcal{V}}(x)$ is **positive (negative) definite** and the function itself is positive (negative), then the equilibrium state x_o is **unstable.** $\left(\mathcal{V} \cdot \dot{\mathcal{V}} > 0\right)$

© Prof. Yong-Hwa Park, KAIST

Liapunov Stability Theorems for a System described by $\dot{x} = f(x)$

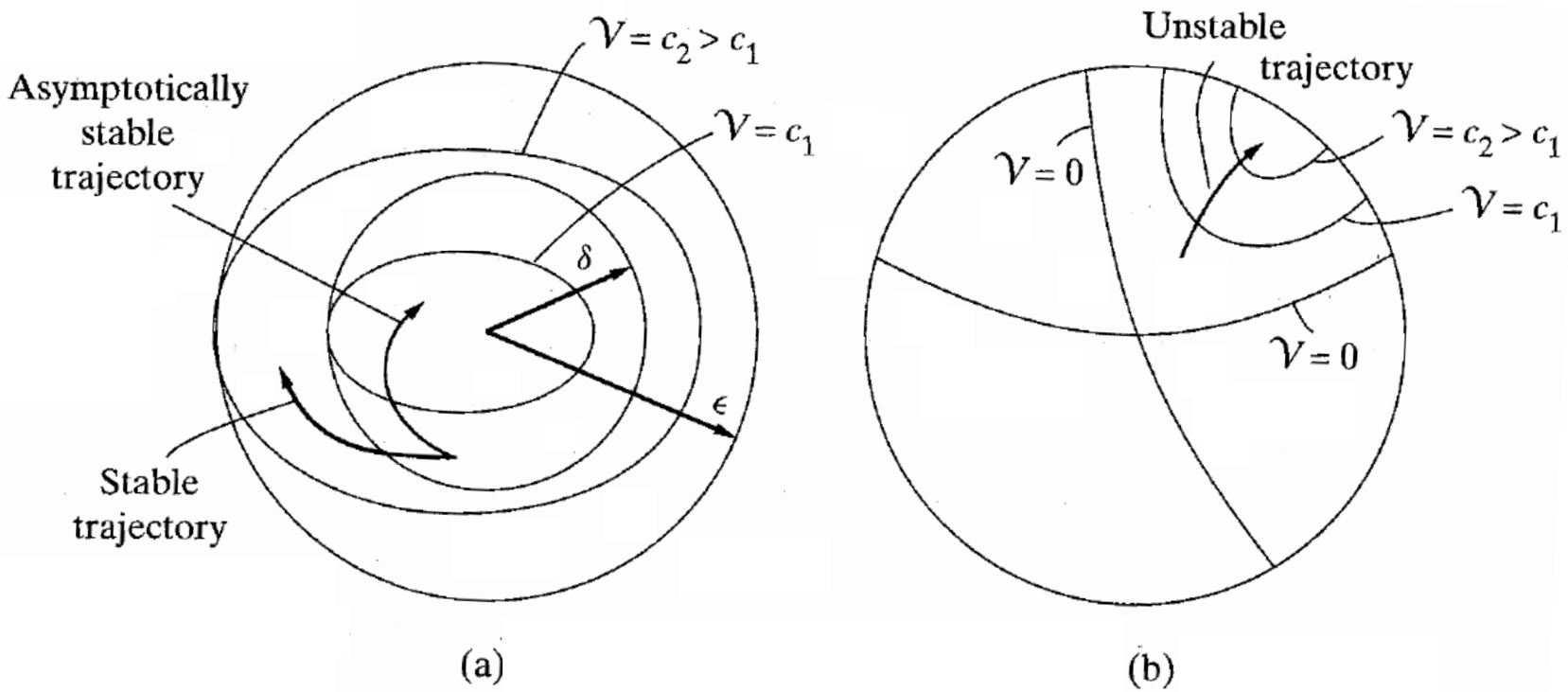


Figure 4.7 **(a)** Geometric interpretation of Liapunov's Theorem 1 (stability) and Theorem 2 (asymptotic stability) **(b)** Geometric interpretation of Liapunov's Theorem 3 (instability)

© Prof. Yong-Hwa Park, KAIST

Stability of a Dynamic System Governed by $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right)-\frac{\partial L}{\partial q}=-\boldsymbol{C}^{*}\dot{q}-\boldsymbol{H}q$

By taking the ***Hamiltonian Function*** as a *Liapunov Function*. Let's try this!

$$\mathcal{H}(p,q)=p^T\dot{q}-L=\dot{q}^T p-L$$

$\downarrow\Leftarrow$ *p*.18, Lecture 8

$$=\underbrace{T_2}_{\text{Positive definite}}+\underbrace{V-T_0}_{U\text{: Sign variable}}$$

q : *generalized coordinate*

$p=\frac{\partial L}{\partial \dot{q}}$: *generalized momenta*

$x=\left[p^T,q^T\right]^T$: *state vector*

$$\dot{\mathcal{H}}(p,q)=\ddot{q}^T p+\dot{q}^T\dot{p}-\left(\left\{\frac{\partial L}{\partial q}\right\}^T\dot{q}+\left\{\frac{\partial L}{\partial \dot{q}}\right\}^T\ddot{q}\right)\overset{p=\left\{\frac{\partial L}{\partial \dot{q}}\right\}}{=}\dot{q}^T\left(\dot{p}-\frac{\partial L}{\partial q}\right)$$

$$\downarrow\Leftarrow\frac{d}{dt}\left(\underbrace{\frac{\partial L}{\partial \dot{q}}}_{=p}\right)-\frac{\partial L}{\partial q}=-\boldsymbol{C}^{*}\dot{q}-\boldsymbol{H}q$$

$$=\underbrace{-\dot{q}^T\boldsymbol{C}^{*}\dot{q}}_{\text{Negative definite}}\ \underbrace{-\dot{q}^T\boldsymbol{H}q}_{\text{Sign variable}}\equiv\dot{\mathcal{H}}(\dot{q},q)$$

When both $\mathcal{H}$ and $d\mathcal{H}/dt$ are sign-variable, *i.e.*, neither Th. 1 nor Th. 2 can be satisfied at a given equilibrium point, then it cannot be stated that the system is stable or unstable at the given equilibrium point.

© Prof. Yong-Hwa Park, KAIST

Stability in Special Cases

$$\mathcal{H}(\dot{q},q) = \underbrace{T_2}_{\text{Positive definite}} + \underbrace{V - T_0}_{U\text{: Sign variable}} \;;\dot{\mathcal{H}}(\dot{q},q) = \underbrace{-\dot{q}^T C^* \dot{q}}_{\text{Negative definite}} \underbrace{-\dot{q}^T H q}_{\text{Sign variable}}$$

$\downarrow \Leftarrow$ No circulartory damping forces: $C' = H = 0\left(C^* = C + C'\right)$

$$\mathcal{H}(\dot{q},q) = T_2 + \underbrace{U}_{\text{Sign variable}} \;;\dot{\mathcal{H}}(\dot{q},q) = \underbrace{-\dot{q}^T C \dot{q}}_{\text{Negative definite}} \quad (\text{in case of 'pervasive' damping})$$

$$\downarrow \Leftarrow \text{ When } \begin{cases} 1) U > 0 \;\left(\text{or } V > T_0\right) \\ 2) T_1 = T_0 = 0 \,(\text{Natural system}) \end{cases}$$

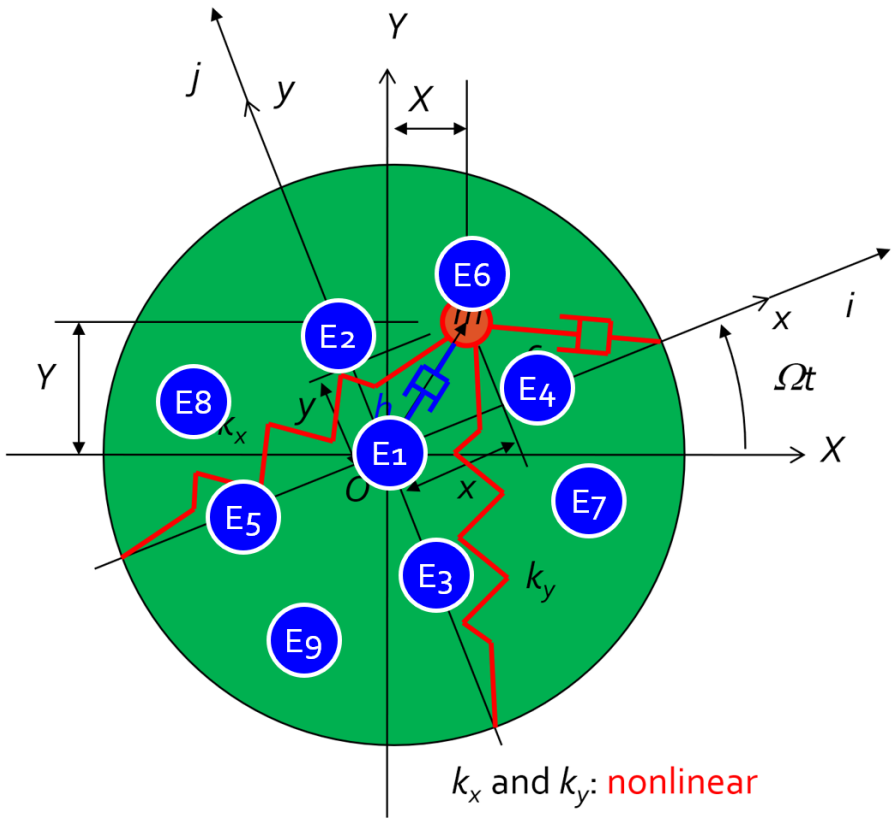
$\mathcal{H}(\dot{q},q)$: positive definite; $\dot{\mathcal{H}}(\dot{q},q)$: negative definte

$\downarrow$

It can be stated that the system is asymptotically stable!

© Prof. Yong-Hwa Park, KAIST

Ex.4.3: Equilibrium points and their stability when h =0 and $m\Omega^2 \geq k_x, m\Omega^2 \geq k_y$ by the Liapunov direct method



9 Equilibrium points:

$$x_0 = 0, x_0 = \pm\sqrt{\left(m\Omega^2 - k_x\right)/k_x\varepsilon_x}$$

$$y_0 = 0, y_0 = \pm\sqrt{\left(m\Omega^2 - k_y\right)/k_y\varepsilon_y}$$

Stability by using Hamiltonian $\mathcal{H}$ as a $\mathcal{V}$:

$$\mathcal{V}(x)\Big|_{x=\begin{Bmatrix} q \\ \dot{q} \end{Bmatrix}} \equiv \mathcal{H}(p,q)\Big|_{p=\frac{\partial L}{\partial \dot{q}}} = \underbrace{T_2}_{\text{Positive definite}} + \underbrace{V - T_0}_{U\text{: Sign variable}}$$

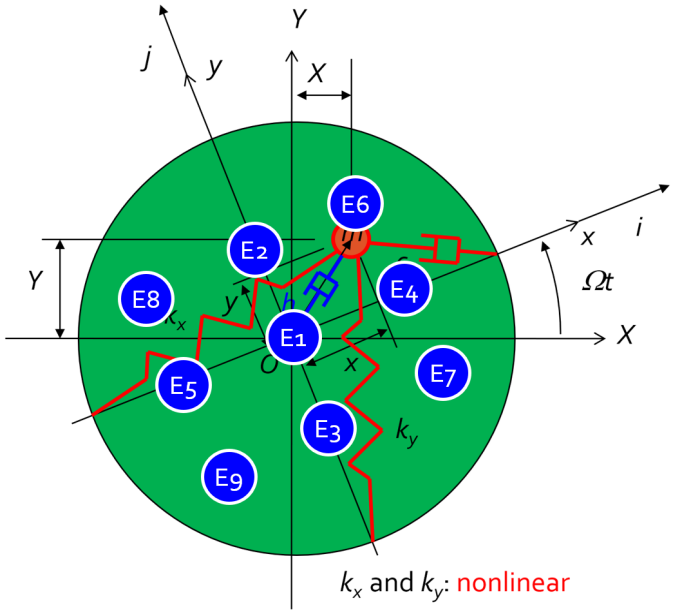
$$\dot{\mathcal{V}}(x)\Big|_{x=\begin{Bmatrix} q \\ \dot{q} \end{Bmatrix}} = \dot{\mathcal{H}}(p,q)\Big|_{p=\frac{\partial L}{\partial \dot{q}}} = \dot{q}^T\left(\dot{p} - \frac{\partial L}{\partial q}\right)$$

$$\downarrow\Leftarrow \begin{cases} \dfrac{d}{dt}\left(\underbrace{\dfrac{\partial L}{\partial \dot{q}}}_{=p}\right) - \dfrac{\partial L}{\partial q} = -\boldsymbol{C}^*\dot{q} - \boldsymbol{H}q \\ h = 0 : \boldsymbol{C}^* = \boldsymbol{C}, \boldsymbol{H} = 0 \end{cases}$$

$$= \underbrace{-\dot{q}^T\boldsymbol{C}\dot{q}}_{\text{Negative definite}}$$

$\therefore$ The stability depends on the sign of U.

© Prof. Yong-Hwa Park, KAIST



$$m\Omega^2 \geq k_x,$$

$$m\Omega^2 \geq k_y$$

Recall equilibrium points (assume h=0):

$$\begin{cases} E_1 : x_0 = y_0 = 0 \\ E_2, E_3 : x_0 = 0, y_0 = \pm\sqrt{\left(m\Omega^2 - k_y\right)/k_y\varepsilon_y} \\ E_4, E_5 : x_0 = \pm\sqrt{\left(m\Omega^2 - k_x\right)/k_x\varepsilon_x}, y_0 = 0 \\ E_6, E_7, E_8, E_9 : x_0 = \pm\sqrt{\left(m\Omega^2 - k_x\right)/k_x\varepsilon_x}, y_0 = \pm\sqrt{\left(m\Omega^2 - k_y\right)/k_y\varepsilon_y} \end{cases}$$

$$\mathcal{V}(x,y,\dot{x},\dot{y}) = \mathcal{H}(x,y,\dot{x},\dot{y}) = T_2 + V - T_0$$

$$\downarrow\Leftarrow \begin{cases} T_2 = \dfrac{m}{2}\left(\dot{x}^2 + \dot{y}^2\right), T_0 = \dfrac{m}{2}\left(x^2 + y^2\right)\Omega^2 \\ V = \dfrac{k_x}{2}\left(x^2 + \dfrac{\varepsilon_x}{2}x^4\right) + \dfrac{k_y}{2}\left(y^2 + \dfrac{\varepsilon_y}{2}y^4\right) \end{cases}$$

$$= \frac{m}{2}\left(\dot{x}^2 + \dot{y}^2\right) + \frac{1}{2}\left\{\left(k_x - m\Omega^2\right)x^2 + \left(k_y - m\Omega^2\right)y^2\right\} + \frac{1}{4}\left(k_x\varepsilon_x x^4 + k_y\varepsilon_y y^4\right)$$

$\dot{\mathcal{V}}(x,y,\dot{x},\dot{y}) = -c\dot{x}^2 < 0$: negative definite

$\therefore$ As long as a positive definite $\mathcal{V}$ or $(V - T_0)$ is available, the system is asymptotically stable!

$$\mathcal{V}(x,y,\dot{x},\dot{y}) = \frac{m}{2}\left(\dot{x}^2 + \dot{y}^2\right) + \frac{1}{2}\left\{\left(k_x - m\Omega^2\right)x^2 + \left(k_y - m\Omega^2\right)y^2\right\} + \frac{1}{4}\left(k_x\varepsilon_x x^4 + k_y\varepsilon_y y^4\right)$$

$$\downarrow\Leftarrow \left\{x(t) = x_0 + x_1(t), y(t) = y_0 + y_1(t),\right.$$

$$\mathcal{V}(x,y,\dot{x},\dot{y})\Big|_{E_1: x_0=0, y_0=0} = \frac{m}{2}\left(\dot{x}_1^2 + \dot{y}_1^2\right) + \frac{1}{2}\left\{\underbrace{\left(k_x - m\Omega^2\right)}_{\text{negative}}x_1^2 + \underbrace{\left(k_y - m\Omega^2\right)}_{\text{negative}}y_1^2\right\} + \frac{1}{4}\left(k_x\varepsilon_x x_1^4 + k_y\varepsilon_y y_1^4\right)$$

$\Rightarrow$ can be negative: unstable

$$\mathcal{V}(x,y,\dot{x},\dot{y})\Big|_{E_2\&E_3: x_0=0, y_0=\pm\sqrt{\left(m\Omega^2-k_y\right)/k_y\varepsilon_y}} = \frac{m}{2}\left(\dot{x}_1^2 + \dot{y}_1^2\right) + \frac{1}{2}\left\{\underbrace{\left(k_x - m\Omega^2\right)}_{\text{negative}}x_1^2 + \frac{1}{2}k_x\varepsilon_x x_1^4\right\} + \frac{1}{4}k_y\varepsilon_y\left(2y_0 + y_1\right)^2 y_1^2$$

$\Rightarrow$ can be negative: unstable

$$\mathcal{V}(x,y,\dot{x},\dot{y})\Big|_{E_4\&E_5: x_0=\pm\sqrt{\left(m\Omega^2-k_x\right)/k_x\varepsilon_x}, y_0=0}$$ $\Rightarrow$ can be negative similarly: unstable

$$\mathcal{V}(x,y,\dot{x},\dot{y})\Big|_{E_6,E_7,E_8,E_9: x_0=\pm\sqrt{\left(m\Omega^2-k_x\right)/k_x\varepsilon_x}, y_0=\pm\sqrt{\left(m\Omega^2-k_y\right)/k_y\varepsilon_y}} = \underbrace{\frac{m}{2}\left(\dot{x}_1^2 + \dot{y}_1^2\right)}_{\text{correct m/4 in text}} + \frac{1}{4}\underbrace{\left\{k_x\varepsilon_x\left(2x_0 + x_1\right)^2 x_1^2 + k_y\varepsilon_y\left(2y_0 + y_1\right)^2 y_1^2\right\}}_{\text{positive}}$$

$\Rightarrow$ positive definite : asymptotically stable

© Prof. Yong-Hwa Park, KAIST

4.5 Stability of Linearized Systems

From Lecture 9: Linearized EOM

$$M\ddot{q}+\left(C^{*}+G\right)\dot{q}+\left(K+H\right)q\left(t\right)=0:$$

$$\ddot{q}=-M^{-1}\left(C^{*}+G\right)\dot{q}-M^{-1}\left(K+H\right)q$$

$$\downarrow\Leftarrow \text{Augmentation by } \dot{q}=\dot{q}$$

$$\underbrace{\frac{d}{dt}\begin{Bmatrix} q \\ \dot{q} \end{Bmatrix}}_{\dot{x}}=\underbrace{\begin{bmatrix} 0 & I \\ -M^{-1}\left(K+H\right) & -M^{-1}\left(C^{*}+G\right) \end{bmatrix}}_{A:\ (2n\times 2n)\ \text{unsymmetric constant matrix}}\underbrace{\begin{Bmatrix} q \\ \dot{q} \end{Bmatrix}}_{x:\ \text{state vector}}:$$

$$\dot{x}=Ax$$

$$\downarrow\Leftarrow x=Xe^{\lambda t}$$

$$AX=\lambda X:\ \text{Algebraic eigenvalue problem}$$

$$\downarrow\Leftarrow \begin{cases} \lambda_j=\alpha_j+i\beta_j, X_j, j=1,2,\ldots,2n \\ x_j=X_je^{(\alpha_j+i\beta_j)t}=X_je^{\alpha_jt}(\cos\beta_jt+i\sin\beta_jt) \end{cases}$$

We can find the stability characteristics of the system in a systematic way by eigenvalues!

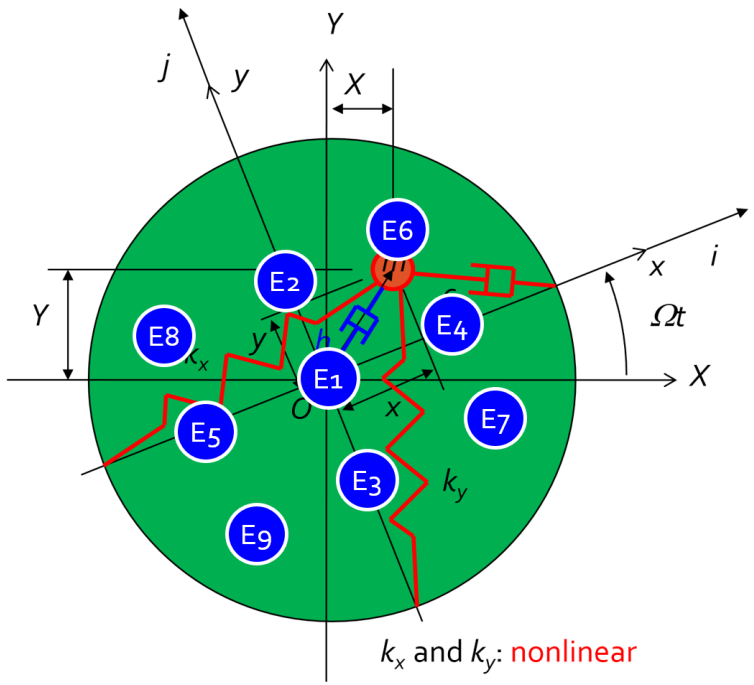
$$x\left(t\right)=\sum_{j=1}^{2n}X_je^{\lambda_jt}\begin{cases} \text{all } \alpha_j<0:\ \text{asymtotically stable} \\ \alpha_j=0:\ \text{stable} \\ \text{at least one } \alpha_j>0:\ \text{unstable} \end{cases}$$

The real part of the eigenvalues controls the stability characteristics of the system !

© Prof. Yong-Hwa Park, KAIST

Ex. 4.5 Stability of Linearized Equations

when $\left(m\Omega^2 - k_x\right) > 0$ and $\left(m\Omega^2 - k_x\right) > 0$



Recall equilibrium points (assume *h*=0):

$$\begin{cases} E_1 : x_0 = y_0 = 0 \\ E_2, E_3 : x_0 = 0, y_0 = \pm\sqrt{\left(m\Omega^2 - k_y\right)/k_y\varepsilon_y} \\ E_4, E_5 : x_0 = \pm\sqrt{\left(m\Omega^2 - k_x\right)/k_x\varepsilon_x}, y_0 = 0 \\ E_6, E_7, E_8, E_9 : x_0 = \pm\sqrt{\left(m\Omega^2 - k_x\right)/k_x\varepsilon_x}, y_0 = \pm\sqrt{\left(m\Omega^2 - k_y\right)/k_y\varepsilon_y} \end{cases}$$

Recall linearized equations:

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \end{Bmatrix} + \begin{bmatrix} c+h & -2m\Omega \\ 2m\Omega & h \end{bmatrix}\begin{Bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{Bmatrix} + \begin{bmatrix} k_x - m\Omega^2 & -h\Omega \\ h\Omega & k_y - m\Omega^2 \end{bmatrix}\begin{Bmatrix} x_1 \\ y_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \text{ at } E_1$$

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \end{Bmatrix} + \begin{bmatrix} c+h & -2m\Omega \\ 2m\Omega & h \end{bmatrix}\begin{Bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{Bmatrix} + \begin{bmatrix} k_x - m\Omega^2 & -h\Omega \\ h\Omega & 2\left(m\Omega^2 - k_y\right) \end{bmatrix}\begin{Bmatrix} x_1 \\ y_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \text{at } E_2, E_3$$

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \end{Bmatrix} + \begin{bmatrix} c+h & -2m\Omega \\ 2m\Omega & h \end{bmatrix}\begin{Bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{Bmatrix} + \begin{bmatrix} 2\left(m\Omega^2 - k_x\right) & -h\Omega \\ h\Omega & k_y - m\Omega^2 \end{bmatrix}\begin{Bmatrix} x_1 \\ y_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \text{at } E_4, E_5$$

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \end{Bmatrix} + \begin{bmatrix} c+h & -2m\Omega \\ 2m\Omega & h \end{bmatrix}\begin{Bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{Bmatrix} + \begin{bmatrix} 2\left(m\Omega^2 - k_x\right) & -h\Omega \\ h\Omega & 2\left(m\Omega^2 - k_y\right) \end{bmatrix}\begin{Bmatrix} x_1 \\ y_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \text{at } E_6, E_7, E_8, E_9$$

© Prof. Yong-Hwa Park, KAIST

By investigating eigenvalues (in text)

$$Ax = \lambda x, \text{ where } \begin{cases} A = \begin{bmatrix} 0 & I \\ -M^{-1}(K+H) & -M^{-1}(C^{*}+G) \end{bmatrix} \\ M = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}, K = \begin{bmatrix} k_x - m\Omega^2 & 0 \\ 0 & k_y - m\Omega^2 \end{bmatrix} \ldots, H = \begin{bmatrix} 0 & -h\Omega \\ h\Omega & 0 \end{bmatrix}, C^{*} = \begin{bmatrix} c+h & 0 \\ 0 & h \end{bmatrix}, G = \begin{bmatrix} 0 & -2m\Omega \\ 2m\Omega & 0 \end{bmatrix} \end{cases}$$

$\downarrow \Leftarrow m = 1$ kg, $\Omega = 2$ rad/s, $c = 0.2$ Ns/m, $k_x = k_y = 3$ N/m

Sec. 4.6 Linear Conservative Natural Systems. The Symmetric Eigenvalue Problem 171

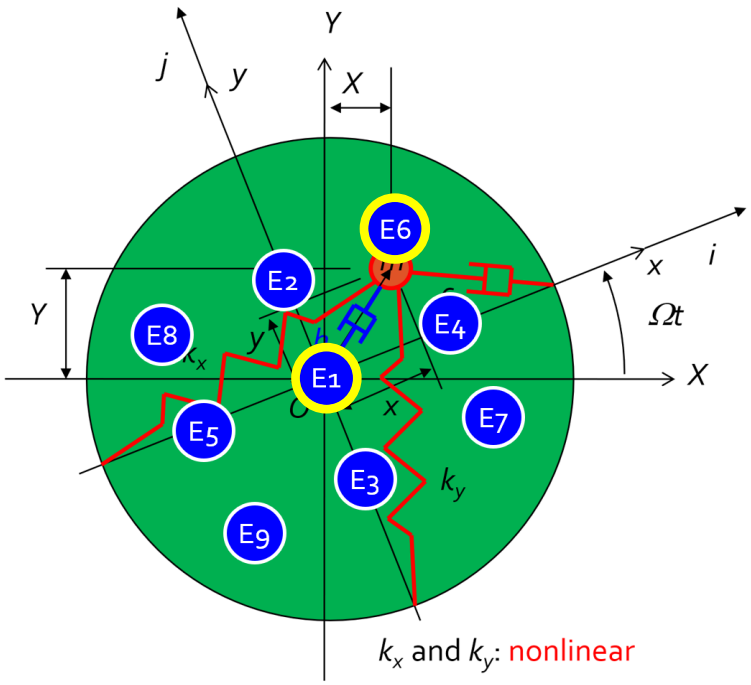
equilibrium points are as follows:

Unstable!	E_1:	$\lambda_1 = \boxed{0.0077} + 0.2678i$	$\lambda_3 = -0.1077 + 3.7309i$	
		$\lambda_2 = \boxed{0.0077} - 0.2678i$	$\lambda_4 = -0.1077 - 3.7309i$	
Unstable!	E_2, E_3:	$\lambda_1 = \boxed{0.3298}$	$\lambda_3 = -0.0877 + 4.1358i$	
		$\lambda_2 = -0.3544$	$\lambda_4 = -0.0877 - 4.1358i$	(m)
Unstable!	E_4, E_5:	$\lambda_1 = -0.3367$	$\lambda_3 = -0.1051 + 4.1362i$	
		$\lambda_2 = \boxed{0.3470}$	$\lambda_4 = -0.1051 - 4.1362i$	
Asymptotically stable!	E_6, E_7, E_8, E_9:	$\lambda_1 = \boxed{-0.0092} + 0.4494i$	$\lambda_3 = \boxed{-0.0908} + 4.4482i$	
		$\lambda_2 = \boxed{-0.0092} - 0.4494i$	$\lambda_4 = \boxed{-0.0908} - 4.4482i$	

Consistent with those by Liapunov direct method on nonlinear EOM in page 12! (page 162 in text)

© Prof. Yong-Hwa Park, KAIST

Stability at E1 and E6 by Liapunov Direct Method
(In-situ quiz of lecture 10)



when $\left(m\Omega^2 - k_x\right) > 0$ and $\left(m\Omega^2 - k_x\right) > 0$

Recall equilibrium points (assume *h*=0):

$$\begin{cases} E_1 : x_0 = y_0 = 0 \\ E_6 : x_0 = \sqrt{\left(m\Omega^2 - k_x\right)/k_x\varepsilon_x}, y_0 = \sqrt{\left(m\Omega^2 - k_y\right)/k_y\varepsilon_y} \end{cases}$$

Perturbation:

$$\begin{cases} x = x_0 + x_1 \\ y = y_0 + y_1 \end{cases}$$

Recall linearized equations:

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \end{Bmatrix} + \begin{bmatrix} c+h & -2m\Omega \\ 2m\Omega & h \end{bmatrix}\begin{Bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{Bmatrix} + \begin{bmatrix} k_x - m\Omega^2 & -h\Omega \\ h\Omega & k_y - m\Omega^2 \end{bmatrix}\begin{Bmatrix} x_1 \\ y_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \text{ at } E_1 \quad \ldots(1)$$

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \end{Bmatrix} + \begin{bmatrix} c+h & -2m\Omega \\ 2m\Omega & h \end{bmatrix}\begin{Bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{Bmatrix} + \begin{bmatrix} 2\left(m\Omega^2 - k_x\right) & -h\Omega \\ h\Omega & 2\left(m\Omega^2 - k_y\right) \end{bmatrix}\begin{Bmatrix} x_1 \\ y_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \text{at } E_6 \quad \ldots(2)$$

In-situ Quiz of Lecture 10: Define *Liapunov* Function(s) of linearized system (1) and (2) on your choice and characterize the stabilities about equilibrium points E1 and E6. Compare your result with the result investigating eigenvalues in page 15. Send your answer to open Q&A in CLASSUM!

© Prof. Yong-Hwa Park, KAIST

Remark: Classification of Linearized Systems

$$M\ddot{q} + \left(\underbrace{C^*}_{\text{sym}} + \underbrace{G}_{\text{skew sym}} \right)\dot{q} + \left(\underbrace{K}_{\text{sym}} + \underbrace{H}_{\text{skew sym}} \right) q(t) = 0$$

$$\text{or } \frac{d}{dt}\begin{Bmatrix} q \\ \dot{q} \end{Bmatrix} = \underbrace{\begin{bmatrix} 0 & I \\ -M^{-1}(K+H) & -M^{-1}(C^*+G) \end{bmatrix}}_{\equiv A\text{: transition matrix}} \begin{Bmatrix} q \\ \dot{q} \end{Bmatrix}$$

$$\downarrow$$

$$\begin{cases} 1)\ \text{Natural conservative system: } C^* = G = H = 0 \Rightarrow M\ddot{q} + Kq = 0 \\ 2)\ \text{Gyroscopic conservative system: } C^* = H = 0 \Rightarrow M\ddot{q} + G\dot{q} + Kq = 0 \\ 3)\ \text{Nonconservative system: } C^* \neq 0 \ \textit{or}\ H \neq 0 \end{cases}$$

Note: although eigenvalue analysis on the transition matrix A can be applied to all of the above three systems, classifications are meaningful for physical understanding & computational considerations!

© Prof. Yong-Hwa Park, KAIST

Highlight 10

- Depending on the stability characteristic around an equilibrium state, we have stable, asymptotically stable and unstable behavior of the motion.

- Liapunov function is a scalar function defined by the state vector that can characterize the stability by considering three Liapunove Theorems. But the Liapunov function is not unique, and we cannot guarantee its existence.

- The stability of linearized motion around an equilibrium state can be systematically examined by the eigenvalue of the transition matrix of the linearized system.

© Prof. Yong-Hwa Park, KAIST